Among states with a $7.25 minimum wage in January 2010, what is the causal effect of later state minimum-wage increases on employment among teenagers ages 16-19?

Data

Data and Sample

I use individual-level data from the IPUMS Current Population Survey (CPS) Basic Monthly samples from January 2010 through December 2019 (Flood et al. 2025). The CPS is a monthly household survey conducted by the U.S. Census Bureau for the Bureau of Labor Statistics. I use all twelve monthly samples in each year so that each state has a larger sample of teenagers.

I restrict the sample to respondents ages 16 through 19 and exclude respondents in the Armed Forces because this study focuses on civilian employment. The CPS provides each respondent's age, sex, state of residence, employment status, and survey weight. I use the CPS Final Basic Weight (WTFINL) when calculating the descriptive statistics in Table 1 and the employment rates in Figure 1.

I combine the CPS with monthly minimum-wage data from Vaghul and Zipperer (2022). Their state minimum-wage variable uses the higher of the state and federal minimum wage. I merge the two datasets by state, year, and month. All observations in the initial civilian teen sample match to a minimum-wage record.

My original proposal compared states only by whether their minimum wage was above the federal $7.25 rate. That definition is not useful for states that were already above $7.25 at the beginning of the sample. I therefore restrict the analytic sample to states with an effective minimum wage of $7.25 in January 2010. A state becomes treated in the first full month when its effective minimum wage rises above the federal minimum. States that stay at $7.25 through 2019 are the comparison group.

The final analytic sample contains 535,323 person-month observations from 36 states. Fifteen states become treated during the sample period, while 21 states never become treated. Because the CPS interviews some households in more than one month, these observations are not 535,323 unique people. They are monthly observations from repeated CPS samples.

Measures

The dependent variable is Employed. It equals one if the CPS classifies a respondent as “at work” or “has job, not at work last week,” and zero for other civilian employment statuses. This measure asks whether a teenager has a job, not whether the teenager is in the labor force.

The main explanatory variable is treatment status. It equals zero before a state raises its effective minimum wage above the federal minimum and one after that change. Because states make this change in different months, treatment starts at different times across states.

I also include age indicators and a Female indicator as individual-level controls. I keep the control set small because the main source of identification is the timing of state policy changes. Prior work has also used CPS data to study minimum wages and teen employment, including Allegretto, Dube, and Reich (2011). Giuliano (2013) shows that employment responses to minimum-wage changes can differ across teenage workers.

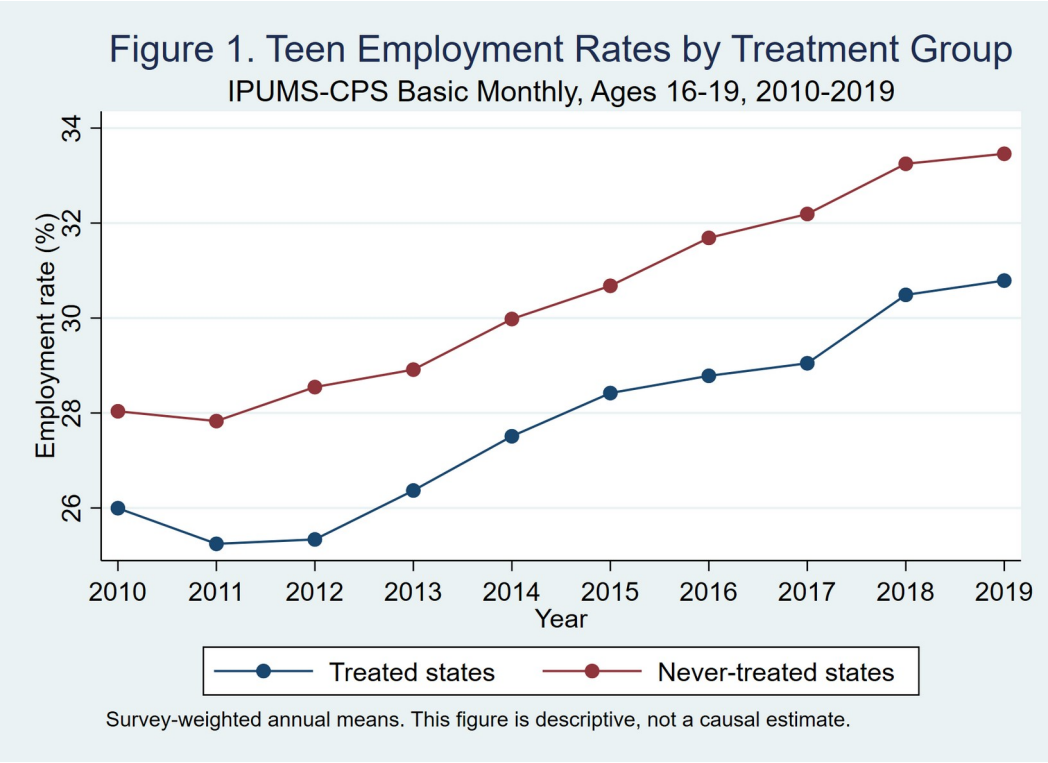
Table 1. Descriptive Statistics

Variable	Analytic sample	Eventually treated states	Never-treated states
Employed (%)	29.40	27.80	30.48
Female (%)	49.22	49.03	49.34
Age (years), average	17.43	17.44	17.43
Effective minimum wage ($), average	7.62	8.18	7.25
Number of observations	535,323	222,819	312,504

Notes: Survey-weighted descriptive statistics for civilian respondents ages 16-19 in the IPUMS-CPS Basic Monthly samples, 2010-2019. The analytic sample includes states with an effective minimum wage of $7.25 in January 2010. Eventually treated states later move above the federal minimum; never-treated states remain at the federal minimum through 2019. Means and percentages use the CPS Final Basic Weight (WTFINL). Minimum-wage data are from Vaghul and Zipperer (2022).

Table 1 shows that 29.4 percent of teenagers in the analytic sample are employed. The employment rate is 27.8 percent in states that eventually become treated and 30.5 percent in never-treated states. The groups are almost identical in average age and gender composition. The average effective minimum wage is higher in the eventually treated states because those states raise their minimum wages during the sample period.

Figure 1 shows that teen employment rises over most of the sample period in both groups. The eventually treated states have lower raw employment rates than the never-treated states throughout the period. This figure is descriptive only. It does not show the causal effect of a minimum-wage increase because treatment begins at different dates across states.



Empirical Approach

I plan to use a staggered difference-in-differences (DiD) design. DiD compares how an outcome changes over time in a treated group with how it changes in a comparison group. In this project, I compare changes in teen employment after a state raises its effective minimum wage with changes in states that remain at the federal minimum. The design is staggered because treated states raise their minimum wages at different times.

Callaway and Sant'Anna (2021) develop a DiD method for settings where treatment begins at different times. They also apply the method to minimum wages and teen employment. Because staggered DiD has not yet been covered in class, this is my current planned approach, and I expect to refine the exact estimation based on feedback.

A simple way to show the variables in the model is:

$$Employment_{ist} = \alpha_s + \lambda_t + \beta\ \text{Treatment}_{st} + \mathbf{X}'_{ist}\gamma + \varepsilon_{ist}. \quad (1)$$

In Equation (1), Employment is the employment outcome for teenager i in state s and month t. Treatment equals one after a treated state raises its effective minimum wage above the federal minimum. The state fixed effects, αs, account for persistent differences across states. The time fixed effects, λt, account for changes that affect all states in the same month. X contains the individual controls for age and sex. Equation (1) is a simple way to show the main parts of the model; the staggered DiD method will allow treatment to begin at different times across states.

The main identifying assumption is parallel trends. In simple terms, this means that without a minimum-wage increase, teen employment in treated states would have changed in a similar way to teen employment in the comparison states. The two groups do not need to have the same employment rate. Figure 1 shows that their levels differ. What matters is whether their trends would have moved similarly if treatment had not occurred.

I plan to use an event-study plot to look at employment trends before and after treatment. The pre-treatment estimates will help me check whether treated states were already moving differently before the policy change. If the pre-treatment estimates are close to zero, that would be consistent with the parallel-trends assumption. Large pre-treatment differences would be a warning that the assumption may not hold.

I plan to cluster standard errors at the state level because the policy changes at the state level and observations within the same state may be related. I also plan to use CPS person weights so the estimates better represent the teen population in the states included in the analytic sample.

References Cited

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